
Final Report

Introduction to Machine Learning Safety CARLA Autonomous Driving Perception System

Student: Sanjana Munikrishnappa

Matriculation number: 261187

Date: 23.08.2026

Contents

1	Code Repository	2
2	System Description	3
2.1	Provided system description	3
2.2	Key design limitations (by construction)	3
2.3	Architecture & Training	4
2.3.1	Model Architecture	4
2.3.2	Training Configuration	4
2.3.3	Dataset Size and Split	4
2.3.4	Class Distribution	5
2.3.5	Architecture Sharing Across Models	5
2.3.6	Resulting Model Performance	5
3	Operational Design Domain	6
3.1	ODD Gap Analysis	6
3.2	ODD Coverage	7
3.3	ODD Violation Response	7
4	Safety Analysis (STPA)	8
4.1	Control structure	8
4.2	Losses	9
4.3	Hazards	9
4.4	Unsafe Control Actions	9
4.5	Causal Loss Scenarios	10
4.6	Safety Constraints	10
5	Safety Case	11
5.1	Verification	11
5.2	V-1: In-distribution Detection Performance	11
5.3	V-2: Robustness to Input Perturbations	12
5.4	V-3: Calibrated Uncertainty	12
5.5	V-4: Out-of-Distribution Detection	13
5.6	V-5: Safe System Fallback	14
6	Residual Risks and Known Design Limitations	15
6.1	Residual Risks	15
6.2	Known Design Limitations	15
7	Deployment Recommendation	16
A	Additional Material	18

1. Code Repository

GitHub Repository:

<https://github.com/Sanjana-Munikrishnappa/Introduction-to-ML-Safety.git>

Note:

The repository contains all the code and a separate `README.md` file for each exercise. In addition, the models were retrained again for the final evaluation and safety analysis. These retrained models are stored separately in the `ML_Safety_Retrained` folder. Thus, the repository contains both the models trained during the individual exercises and the newly retrained models used for the final evaluation.

2. System Description

Provided system description

An autonomous vehicle (AV) prototype operates in an urban environment. The system is composed of the following components:

1. **Camera:** A single forward-facing RGB camera, rigidly mounted behind the windshield, provides images at 10 Hz. There are *no other perception sensors* (no radar, no LiDAR, no ultrasonic).
2. **Perception models:** Three separate binary classifiers each receive the camera frame and output one binary prediction: *pedestrian present*, *vehicle present*, or *traffic light present*. Each model was trained exclusively on sunny daytime data collected in the CARLA simulator.
3. **Distance oracle:** A separate model estimates the distance to detected pedestrians and vehicles. You may assume this model is perfect for the sake of this course.
4. **Planning module:** The rule-based planner consumes the three perception outputs together with the current vehicle speed and steering angle. Its primary safety-relevant decisions are whether to continue driving, request deceleration, or issue an emergency brake command. Emergency braking is triggered when the pedestrian detector reports a pedestrian within a critical distance, when the vehicle detector indicates a road user ahead within a speed-dependent threshold, or when the traffic-light detector indicates that stopping is required at an approaching intersection. The planner does not have access to raw camera images and relies entirely on the perception models' outputs.
5. **Vehicle actuators:** Throttle, brake, and steering commands are executed by drive-by-wire actuators. Emergency braking applies maximum deceleration within the physical limits of the vehicle.
6. **Human safety operator:** A trained operator sits in the driver seat and monitors the road. The operator can override the autopilot at any time by pressing the brake pedal or turning the steering wheel. The operator receives a visual dashboard showing the current perception output and system status, but no auditory alerts are provided. During testing, operators work 4-hour shifts.
7. **Operating environment:** The vehicle is tested on public urban roads at speeds up to 50 km/h. The intended conditions are daytime, dry weather, and mapped intersections. However, weather and lighting conditions can change during a test drive (e.g. sudden cloud cover, low sun angle, rain onset).

Key design limitations (by construction)

- No sensor redundancy — a single camera is the only perception input.
- No other safety mechanisms are deployed yet.
- The human operator is the only fallback if the automation fails.
- The operator may exhibit a non-zero probability of delayed or missed intervention, especially under prolonged monitoring.

Architecture & Training

2.3.1 Model Architecture

The three perception models the traffic light detector, pedestrian detector, and vehicle detector share an identical common custom CNN backbone called **TrafficLightClassifier**. The network is trained from scratch and consists of a feature extractor followed by a classification head.

Feature Extractor : The feature extractor consists of three convolutional blocks:

- Conv2D(3 -> 16 channels, 3x3 kernel, padding=1) followed by ReLU activation and MaxPool2D(2x2).
- Conv2D(16 -> 32 channels, 3x3 kernel, padding=1) followed by ReLU activation and MaxPool2D(2x2).
- Conv2D(32 -> 64 channels, 3x3 kernel, padding=1) followed by ReLU activation and MaxPool2D(2x2).

Classification Head : After feature extraction, the resulting feature representation is flattened and passed through two fully connected layers:

- Flatten
- Linear(50,176 -> 128) followed by ReLU activation
- Linear(128 -> 1)

The input images are resized to $224 \times 224 \times 3$ RGB images before being provided to the network. The network produces a single logit corresponding to the presence (1) or absence (0) of the target object class in the image. Each detector uses the same architecture, while the training target differs between traffic lights, pedestrians, and vehicles.

2.3.2 Training Configuration

Parameter	Value
Optimizer	Adam
Learning rate	$1 \times e^{-3}$
Loss function	BCEWithLogitsLoss (binary crossentropy with logits)
Epochs	3
Batch size	32
Input resolution	$224 \times 224 \times 3$
Pretrained weights	None

2.3.3 Dataset Size and Split

The dataset consists of CARLA simulation imagery and contains a total of 14,400 images. The dataset is divided into training, validation, and test sets as follows:

Split	Samples	Proportion
Train	7,200	50%
Validation	3,600	25%
Test	3,600	25%
Total	14,400	100%

2.3.4 Class Distribution

The class distribution differs between the three binary classification tasks. In particular, the pedestrian dataset is substantially more imbalanced than the other classes.

Label	Positive	Negative
has_traffic_light	5,276	1,924
has_pedestrian	1,718	5,482
has_vehicle	5,458	1,742

2.3.5 Architecture Sharing Across Models

Each of the three models uses exactly the same architecture, loss function, optimizer type, learning rate, batch size, and number of training epochs. The only difference between the models is the training target label:

- `has_traffic_light` for the traffic light detector,
- `has_pedestrian` for the pedestrian detector,
- `has_vehicle` for the vehicle detector.

Consequently, the three detectors can be considered separate binary classifiers that share the same model design while being trained independently for their respective target classes.

2.3.6 Resulting Model Performance

The resulting validation performance after three training epochs is summarized below.

Model	Final Training Loss	Validation Accuracy
Traffic Light Detector	0.1210	95.94%
Pedestrian Detector	0.4747	73.47%
Vehicle Detector	0.3793	74.83%

The traffic light detector achieves the highest validation accuracy at 95.94%. In contrast, the pedestrian detector achieves the lowest validation accuracy at 73.47%. The pedestrian detector also operates on the most imbalanced class distribution, with 1,718 positive samples compared with 5,482 negative samples.

This difference is directly relevant to the safety analysis. Reduced pedestrian detection performance can increase the likelihood of an undesirable control action, such as failing to detect a pedestrian. Such a perception failure can result in an unsafe control action, which may lead to a vehicle hazard and ultimately to a loss involving pedestrian injury.

3. Operational Design Domain

The models were trained on CARLA simulation clear/sunny daytime imagery of a single town map, using a front-facing RGB camera with an input resolution of 224×224 .

Coverage of training data:

All training, validation, and test data were collected under sunny daytime conditions on a single town map. No fog, night-time conditions, or alternative towns were encountered during training.

Dimension	Operating conditions	Non-operating conditions	Detection
Weather	Clear/sunny	Fog	Confidence drops from 0.70 to 0.53 for traffic lights; k-NN AUROC = 1.0, MSP AUROC = 0.71
Lighting	Daytime	Night	Confidence collapses near zero (e.g. 0.0008 for traffic lights); k-NN AUROC = 1.0, MSP AUROC = 1.0
Camera	Front-facing RGB, fixed mount	Other camera angle/sensor	Not evaluated
Scene type	Trained town map	Different town	k-NN AUROC = 1.0, MSP AUROC = 0.74
Speed range	Not specified in training data	Not empirically tested	No empirical evidence available

ODD Gap Analysis

1. ODD dimensions not covered by training data:

- Weather : training data is sunny only; fog is untested during training.
- Lighting : daytime only; no night or low-light training examples.
- Scene/town: single town map only; no geographic diversity.
- Camera angle/sensor: only one fixed front-facing RGB camera; no other viewpoints or sensor types.
- Speed range: never recorded or varied in the dataset.

2. Conditions where no performance claim can be made:

- Night driving: confidence collapses near zero for all three models.
- Any other town/map beyond the one tested: only one alternate town was probed; performance on other unseen layouts is unknown
- Camera failure, occlusion, or alternate mounting configurations.
- Any speed-dependent behaviour.

ODD Coverage

This is a combinatorial (k -way) coverage metric, not one that the notebooks would calculate out of the box, but it can be computed from the ODD tests as written, i.e., from Example 7, where the OOD tests use a one-at-a-time change from Sunny, Day, Train Town as the baseline.

We use the three varied dimensions:

- Weather
- Lighting
- Town

Camera and speed are kept fixed, as there is no variation in these dimensions.

Evidence (k -projection coverage):

k	Observed	Possible	Coverage
1	6	6	100%
2	9	12	75%
3	4	8	50%

ODD Violation Response

V-4 – ODD violation detection (verified):

K-NN distance-based OOD detection achieves an AUROC of 1.0 across Fog, Night, and Different Town conditions. MSP/confidence-based methods perform more weakly. Therefore, V-4 holds, indicating an ability to detect deviation from the intended ODD.

V-5 – Response to identified ODD violation (argued rather than validated):

The only ODD violation response policy specified is the safety constraint that if model confidence falls below a threshold of 0.6, vehicle velocity should be reduced to 15 km/h. This constraint is based on intended behaviour and has not been implemented or verified within the provided framework. None of the provided notebooks simulate velocity reduction or a default response action following ODD detection.

Accordingly, V-5 should be treated as a design argument rather than an empirically validated claim.

Although V-4 is strongly supported, V-5 is not validated because no execution evidence is available.

4. Safety Analysis (STPA)

Control structure

The safety-relevant control structure consists of two controllers: the automated planning module and the human operator. The planning module receives information from the perception models and generates vehicle control commands. The human operator can override the automated system by taking manual control.

The perception subsystem receives information from the camera and distance oracle. The perception models provide perception information to the controllers, while the vehicle actuators execute steering, throttle, and brake commands.

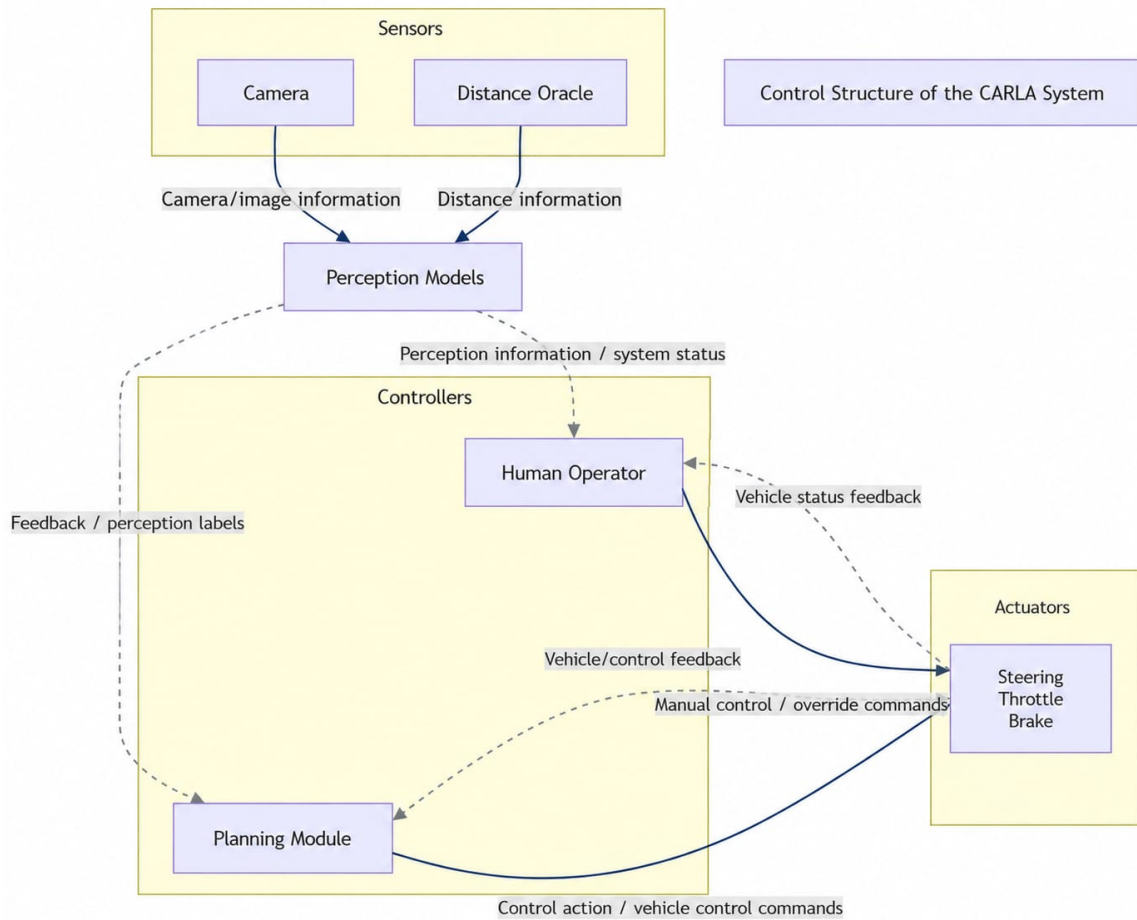


Figure 1: Control structure of the CARLA system.

Losses

ID	Loss	Why unacceptable
L-1	Collision with a pedestrian	Injury or death
L-2	Collision with another vehicle	Injury, damage
L-3	Collision at a red-light intersection	Injury, legal liability
L-4	Loss of human trust/control over the vehicle	Operator cannot intervene in time

Hazards

ID	Hazard	Loss(es)
H-1	Vehicle fails to stop for a pedestrian in its path	L-1
H-2	Vehicle enters intersection against a red light	L-3
H-3	Vehicle fails to react to another vehicle in its path	L-2
H-4	Vehicle acts on unreliable perception output (ODD violation, no fallback)	L-1, L-2, L-3
H-5	Human operator does not regain control in time after a system failure	L-1, L-2, L-3, L-4

Unsafe Control Actions

An Unsafe Control Action (UCA) is a control action that becomes unsafe under a specific condition. It identifies the controller, control action, failure type, and unsafe context, while the reason is explained separately in the loss scenario. The two controllers are the automated planning module and the human operator.

ID	Controller	Control action	UCA type	Hazard	Context in which unsafe (s)
UCA-1	Planning module	Brake command	Not provided	H-1	Pedestrian present, but low light causes detector confidence to collapse (~ 0.002), so no pedestrian is flagged.
UCA-2	Planning module	Brake command	Not provided	H-2	Red light present, but night/fog conditions degrade traffic-light detection confidence.
UCA-3	Planning module	Brake command	Provided incorrectly	H-3	System continues acting on detector output even when operating outside the trained ODD (fog, night, new town).
UCA-4	Human operator	Takeover /override	Too late	H-4	No runtime ODD-violation alert exists, so the operator has no cue to intervene before a failure occurs.
UCA-5	Planning module	Steer/brake command	Provided incorrectly	H-5	Vehicle detection confidence drops in fog.

Causal Loss Scenarios

A loss scenario explains why a UCA occurs—the causal factor that the safety constraint must control.

The identified loss scenarios arise from different causal factors in the perception and control system. These include inputs outside the trained operational domain, overconfident model predictions caused by class imbalance, and adversarial perturbations affecting detection performance.

LS-ID	For UCA	Causal scenario (why the UCA occurs)
LS-1	UCA-1	Undetected out-of-ODD input. Night conditions were never included in training data. Pedestrian confidence collapses to approximately 0.002, so a real pedestrian is not flagged.
LS-2	UCA-1	Overconfident misclassification caused by class imbalance. The pedestrian class is underrepresented in the training data (1,718 positive samples out of 7,200), biasing the model toward negative predictions even in-domain.
LS-3	UCA-3	Undetected out-of-ODD input. The system has no runtime check for fog, night, or new-town conditions, so it continues acting on unreliable detector output without detecting the violation.
LS-4	UCA-2, UCA-5	Adversarial perturbation. An FGSM attack at $\varepsilon = 0.05$ drops traffic-light recall by 64% and vehicle recall by 97%, meaning small crafted perturbations can silently defeat detection.

Safety Constraints

A safety constraint specifies what must be true of the system so that a causal loss scenario cannot lead to an unsafe control action.

SC-ID	Controls (LS-ID)	Safety constraint (requirement)	Level	Verified by
SC-1	LS-1	System must detect out-of-ODD lighting conditions (e.g. night) and not act on unreliable pedestrian confidence.	System	V-3
SC-2	LS-2	Pedestrian training data must be rebalanced or reweighted so recall does not degrade due to class imbalance.	Model	V-1
SC-3	LS-3	System must continuously monitor for ODD violations (fog, night, new town) using a distance/feature-based detector, not raw confidence alone.	System	V-4
SC-4	LS-4	Detection models must maintain recall above a defined threshold under bounded adversarial perturbation ($\varepsilon \leq 0.05$).	Model	V-2
SC-5	LS-3	On detected ODD violation, system must trigger a defined fallback (e.g. speed reduction, operator handover) rather than continuing normal operation.	System	V-5

5. Safety Case

Verification

The safety case verification assesses each safety claim against the relevant safety constraints, loss scenarios, and hazards. The following table provides the overall verification mapping for claims V-1 to V-5.

Verification	Verifies SC(s)	Loss scenario(s)	Hazard(s)
V-1: In-distribution Detection Performance	SC-2	LS-2	H-1
V-2: Robustness to perturbations	SC-4	LS-4	H-2, H-3
V-3: Calibrated uncertainty	SC-1	LS-1, LS-2	H-1
V-4: Out-of-distribution detection	SC-3	LS-3	H-4
V-5: Safe system fallback	SC-5	LS-3	H-4

V-1: In-distribution Detection Performance

Verifies: SC-2 **Hazard(s):** H-1 **Loss scenario(s):** LS-2

Constraint: The model achieves sufficient recall when operating within the ODD.

Metric: Per-class recall on the in-distribution test set.

Evidence (In-Distribution Detection Performance)

Model	Recall	Threshold	Met?
Pedestrian	9.63%	≥ 0.90	No
Traffic light	97.10%	≥ 0.85	Yes
Vehicle	91.33%	≥ 0.85	Yes

Note: 73.47% in Section 2.3.6 is validation accuracy from training, a separate metric — not the V-1 recall result.

Overall Verdict: $\approx$ **Partial**— The pedestrian model achieves only 9.63% recall against the 90% target (Section 5.2), leaving H-1 severely under-controlled. Class rebalancing and a weighted loss function are required.

Supporting Evidence: Explanation Quality

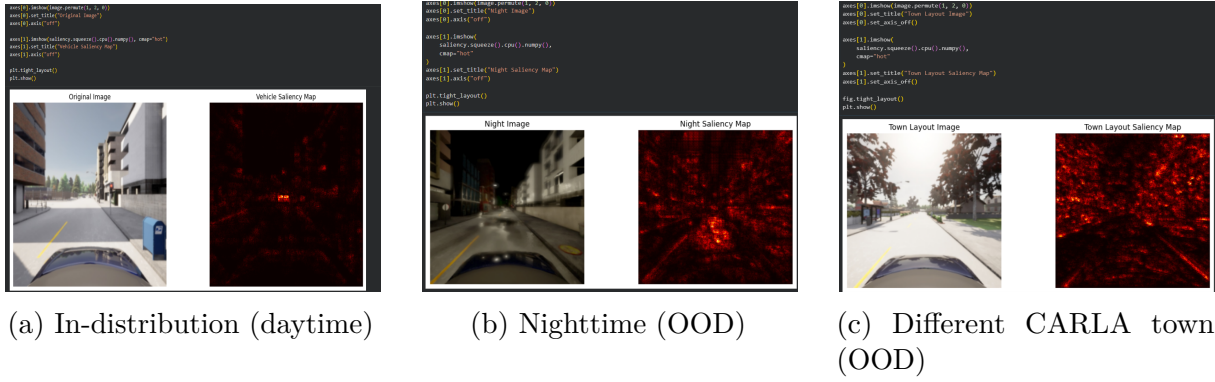


Figure 2: Grad-CAM overlays for the pedestrian detector across conditions.

V-2: Robustness to Input Perturbations

Verifies: SC-4 **Hazard(s):** H-2, H-3 **Loss scenario(s):** LS-4

Constraint: recall does not degrade substantially under small adversarial perturbations, limiting the impact of adversarial manipulation of the camera feed.

Metric: Recall drop under FGSM at $\varepsilon = 0.05$ (clean $\rightarrow$ adversarial).

Evidence (Adversarial Robustness)

Model	Recall (clean)	Recall	Drop	Met?
Pedestrian	4 positives	0 positives	100%	No
Traffic light	65 positives	1 positive	98.5%	No
Vehicle	100 positives	3 positives	97%	No

Overall Verdict: $\times$ **Not met**— FGSM at $\varepsilon = 0.05$ causes recall drops of 100% (pedestrian), 98.5% (traffic light), and 97% (vehicle) (Section 5.3). Adversarial training or certified robustness methods are required. Exercise 8’s recall calculation has a known bug (Appendix A); treat these figures as indicative of severe vulnerability.

Result: At $\varepsilon = 0.05$, all three models show a recall drop well above the 10% threshold. The pedestrian detector loses all positive detections, while the traffic-light and vehicle detectors lose 98.5% and 97% of their positive detections, respectively.

V-3: Calibrated Uncertainty

Verifies: SC-1 **Hazard(s):** H-1 **Loss scenario(s):** LS-1, LS-2

Constraint: The models provide well-calibrated confidence estimates, so low-confidence predictions can trigger fallback behaviour rather than silently overcommitting.

Metric: Expected Calibration Error (ECE) on the in-distribution test set.

Evidence (Uncertainty)

Model	ECE (single)	ECE (temp. scaled)	Met?
Pedestrian	0.4786	0.4932	No
Traffic light	0.2352	0.2352	No
Vehicle	0.1075	0.0363	No

Overall Verdict: × **Not met.** Temperature scaling does not provide sufficient calibration across all three perception models. Therefore, SC-1 is not satisfied.

Result: The pedestrian and traffic-light models remain above the ECE threshold of 0.05 after temperature scaling. Although the vehicle model improves to an ECE of 0.0363, the overall requirement is not satisfied because all three models must provide sufficiently calibrated confidence estimates.

V-4: Out-of-Distribution Detection

Verifies: SC-3 **Hazard(s):** H-4 **Loss scenario(s):** LS-1, LS-3

Constraint: An accompanying monitor reliably flags inputs that fall outside the ODD, preventing silent failure on conditions not covered by training data (fog, night).

Metric: AUROC of the OOD detector separating in-distribution from OOD inputs.

Detector used: k-NN feature-distance (1-nearest-neighbour), compared against MSP (Maximum Softmax Probability) as a baseline.

Evidence (Out-of-Distribution Detection)

Model	Detector	AUROC	Met?
Pedestrian	Not evaluated	—	Not tested
Traffic light	k-NN	1.00	Yes
Traffic light	MSP (baseline)	0.82	No
Vehicle	Not evaluated	—	Not tested

Overall Verdict: ≈ **Partial.** SC-3 is met partially for only one of the three models. The k-NN detector successfully identifies OOD conditions for the traffic-light model, but the evidence does not apply to the overall project because the pedestrian and vehicle models remain untested.

V-5: Safe System Fallback**Verifies:** SC-5 **Hazard(s):** H-4 **Loss scenario(s):** LS-3

Constraint: the system provides a safe fallback when a perception model fails or operates outside the ODD — even if a model produces incorrect output, the architecture limits propagation of the error to harm through fallback behaviour and human oversight.

Evidence (Design argument)

Fall-back behavior defined: When V-4 is predicted as OOD by k-NN detector OR when confidence drops below the predetermined threshold, system will throttle speed down to 15 km/h and signal the human to take over based on Exercise 5 constraint (if model confidence drops below threshold = 0.6, reduce speed to 15 km/h).

Bound to operator takeover: Design assumes the human (the second controller described in the control structure, Figure 1) could indeed retake control within the fallback time without reaching an explicitly time-bound situation. No specific time boundary was stated or analyzed from exercises.

System-level constraint addressed: The SC-3 (OOD monitoring) combined with SC-5 (fallback) makes a clear safety backup.

Overall verdict: × **Not met.**

This is still a design specification. No notebook code simulates, or tests the fallback mechanism, speed reduction, or the operator transfer. There is no indication on what the real system does, if it has been triggered, nor a defined, and verified time bound upon operator takeover. Keep this as an open item, and place into section 6 (residual risks).

6. Residual Risks and Known Design Limitations

Residual Risks

V.1 – In-distribution accuracy – Not met.

The pedestrian model achieves only 73.47% accuracy, below the 90% target. The pedestrian class is imbalanced (1718:5482), leaving H-1 severely under-controlled. Class balancing or a weighted loss function is required.

V.2 – Robustness to perturbations – Not met.

An FGSM attack with $\varepsilon = 0.05$ caused approximately 97% loss in output performance for all three models. Adversarial training or certified robustness methods are required.

V.3 – Calibrated uncertainty – Not met.

The ECE of all three models remains above the acceptable threshold of 0.05. For the pedestrian model, temperature scaling worsened ECE from 0.479 to 0.493. Class-specific calibration methods are required.

V.4 – Out-of-distribution detection – Partial.

The k-NN OOD detection achieved AUROC = 1.0 for the traffic-light model, but was not tested for the pedestrian and vehicle models. Evaluation should be extended to these models.

V.5 – Safe system fallback – Not met.

The proposed fallback consists of speed reduction and human handover, but has not been simulated or evaluated with operator response time. This must be implemented and tested.

Known Design Limitations

1. Presence-only traffic light detection.

The traffic-light model only detects whether a traffic light is present; it cannot classify red, yellow, or green. Therefore, H-2 remains possible.

2. Distance oracle instead of a real sensor.

The distance oracle provides simulation ground truth rather than measurements from LiDAR, radar, or stereo cameras. Real-world distance estimation errors are therefore not addressed.

3. No sensor redundancy.

All detection models use a single forward-facing RGB camera. If the camera is occluded or damaged, the system loses external perception information.

4. Uncoordinated detection outputs.

The pedestrian, vehicle, and traffic-light models operate independently and cannot identify inconsistencies between their outputs, limiting detection of certain perception failures.

7. Deployment Recommendation

Recommendation : Do not deploy

Conditions / restrictions :

Not applicable — system does not meet minimum safety evidence for deployment in any configuration at this stage.

Justification :

The safety case is negative. Only V-4 is partially fulfilled; none of the five verifications are fully met. V-1 fails outright: the pedestrian model achieves only 9.63% recall against a 90% requirement, meaning it misses the overwhelming majority of pedestrians even within its trained ODD. V-2 fails as well: all three models lose nearly all positive detections under a small adversarial perturbation (100% / 98.5% / 97% recall drop at $\epsilon = 0.05$). The calibration models (V-3) remain above the allowed ECE threshold, and the pedestrian model's calibration worsens after temperature scaling. V-5 states a fallback design but provides no implementation or verification evidence (0/3 constraints discharged).

A system that cannot detect pedestrians reliably, cannot detect when its own detection has failed, and has no verified fallback for that situation represents an intolerable and unaddressed risk to human beings. No deployment — including a conditionally restricted one — should proceed until V-1, V-2, V-3, and V-5 are addressed, at minimum through pedestrian data rebalancing, adversarial robustness improvements, recalibration, and implementation and testing of the fallback mechanism.

Statement of Authorship

I hereby declare that I have independently written the present work and have not used any sources or aids other than those stated.

Texts, trains of thought, concepts, graphics, etc. from external sources that I have directly or indirectly adopted in my work I have clearly marked as such and provided with complete references to the respective source. All other content of this work (text parts, figures, tables, etc.) without corresponding references originates in the copyright sense from me.

I have exclusively used AI systems whose use was approved by the examiner as an aid.

Excluded from the requirement of identification are orthographic or grammatical corrections, translations, and improvements to formulations that do not alter meaning.

I am aware that AI-generated texts offer no guarantee for the quality of content and text. I therefore declare that I have used AI systems only as an aid, that I have listed and marked the AI-generated content, critically reviewed it, and that my independent contribution predominates in this work.

I confirm that I have fully understood the content of my work and can represent it independently.

Magdeburg, 23 August 2026

Place, Date

Sanjana Munikrishnappa

Signature

A. Additional Material

- Exercise 3 — model training notebook: architecture, training curves, dataset split, and class distribution for all three detectors. Contains the initial accuracy results (95.94% / 73.47% / 74.83%) referenced throughout this report.
- Exercise 4 — dataset loading and preprocessing pipeline (train/validation/test extraction, label schema, image transforms) underlying all later exercises.
- Exercise 5 — temperature scaling and confidence threshold analysis, plus a backdoor (data poisoning) attack simulation not otherwise covered in this report’s LS-n scenarios.
- Exercise 6 — saliency map generation for the traffic light model across fog, night, and different-town conditions; source of the evidence used in Figure 2.
- Exercise 7 — ODD coverage analysis (confidence and AUROC-based OOD detection, MSP vs. k-NN); source of the evidence used in V-4.
- Exercise 8 — adversarial robustness evaluation using FGSM at $\epsilon = 0.01, 0.05, 0.10$; source of the evidence used in V-2. Note: contains a known bug in its recall-calculation function, flagged in this report.
- Exercise 9 — calibration analysis (ECE before/after temperature scaling); source of the evidence used in V-3.